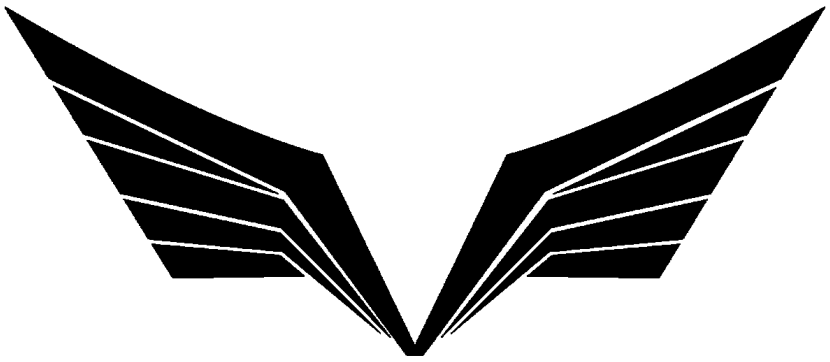


SUN DEVIL
ROCKETRY

Avionics Firmware Development Team

Quality Control Document

Project:	Flight Computer Firmware
Part Number:	L0008-110
Drafted By:	Eli Sells
Project Engineer:	Eli Sells
Quality Assurance:	Allison Cheon
Chief Engineer:	Eli Sells

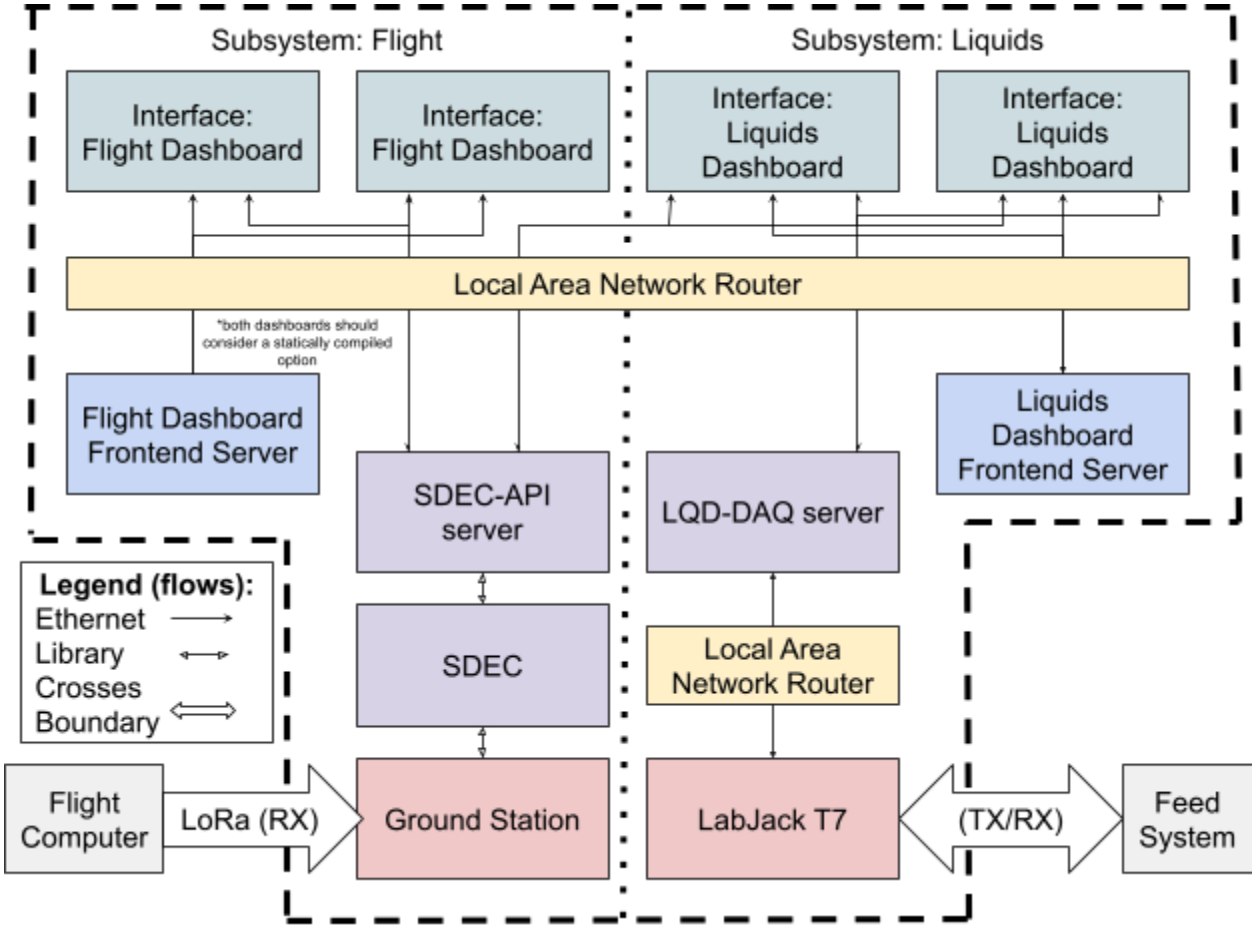
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1. PROJECT OVERVIEW

The New Liquid Feed System is Sun Devil Rocketry's answer to the problems from the last generation. The proprietary hardware required long development times to add/remove sensors, and the data logging rate was incredibly unreliable.

This project is a distributed system with distinct hardware and software components. Contained within the system boundary are the following components:



1. User Interfaces
    a. Flight Dashboard
    b. Liquids Dashboard
2. Hardware Interfaces
    a. SDEC-API + SDEC

   b. LQD-DAQ
3. Hardware
   a. Ground Station
   b. LabJack T7
4. Network Interfaces
   a. Layer 3 network interface like a router or managed switch

1.1 QUALITY ASSURANCE PRACTICES

The primary guiding standard for this project is the SDR Software Quality Assurance Standard. This project is used in flight with the possibility of being used to affect flight dynamics, so it is certified to the highest "Flight Code" standard.

In addition to this minimum standard, each of the following components have agreed upon the following quality assurance practices:

1.1.1. FLIGHT & LIQUIDS DASHBOARDS

The dashboards (Software Division) will employ the following practices:

- Regular scanning for issues/vulnerabilities in dependencies

1.1.2. LQD-DAQ

The LQD-DAQ service will employ the following practices:

- No code review baseline exists. For compliance with the QA standard, a baseline will first be established with a full application review.
- Regular scanning for issues/vulnerabilities in dependencies
- Automated testing of major features

1.1.3. SDEC + SDEC-API

SDEC and SDEC-API will employ the following practices:

- Regular scanning for issues/vulnerabilities in dependencies
- SDEC (library): Automated testing of major features

1.1.4. GROUND STATION

The Ground Station will follow the QA standard with no modifications.

1.1.5. FULL SYSTEM

The system as a whole will employ the following practices:

- Additional SQA audits of component level compliance by internal personnel.

2. SUPPORTING DOCUMENTATION

This section shall reference any documentation for the controlled article. This could be anything from formal design documents and procedural documentation to informal user's guides associated with the article.

DOCUMENT	DESCRIPTION
NLFS Functional Hazard Assessment	A hazard analysis and list of mitigations for the firmware.

3. REQUIREMENTS

The article's requirements are defined in the NLFS SysRD.

4. DEVELOPMENT READINESS

The project meets the requirements to begin development:

1. Requirements are defined
2. Quality assurance practices have been defined and agreed upon

Date: 9/2/2026
Project Engineer: Eli Sells
Quality Assurance: Allison Cheon

5. COMPONENTS

This section will provide a breakdown of this component's constituent parts.

"Separately Controlled Components" have their own part number and quality control processes. External/COTS components fall into this category if the supplier applies their own QA processes, although acceptance testing is encouraged.

"Components with Combined Control" get their quality control coverage from this project. Each component in this table is expected to have some requirement and test criteria later in the document.

5.1. SEPARATELY CONTROLLED COMPONENTS

COMPONENT	PART NUMBER & VERSION	DESCRIPTION	QC STATUS
LabJack T7	P/N: T7	A COTS platform for data acquisition and control.	External

5.2. COMPONENTS WITH COMBINED CONTROL

COMPONENT	DESCRIPTION	RELEVANT REQUIREMENTS	RELEVANT TEST PROCEDURES
LQD-DAQ	Python service that provides APIs to operate the test cart. *(SHA: db83ff20b3b7cd1eba2deb5b6a4d7e2f61b6c3f4)*	3.1, 3.1.1, 3.2.1, 3.2.3, 3.2.7, 3.2.8, 3.4, 3.5	TBD
Liquid GUI	Graphical user interface *(SHA: ad044494c301d211d0eede722dcd0f5e0c74bf0b)*	3.1, 3.1.1, 3.2.x, 3.5	TBD

6. TEST PROCEDURES

All test procedures are defined (but not described) below. The full description for each test is in the NLFS Test Plan & Results document. The requirement tags below were designated in NLFS Requirement Tracing following SDR's tagging/tracing convention for safety critical projects. Procedures designated "[project] Unit Test" are defined in code.

Tracing is provided in the NLFS Requirement Tracing spreadsheet.

TEST PROCEDURE	REQUIREMENT(s)	TEST RESULT
Structural Requirement Analysis	RQ.NLFS.00001, RQ.NLFS.00002, RQ.NLFS.00003	TBD
UI System Test	RQ.NLFS.00004, RQ.NLFS.00005, RQ.NLFS.00006, RQ.NLFS.00008, RQ.NLFS.00009, RQ.NLFS.00014, RQ.NLFS.00042, RQ.NLFS.00043, RQ.NLFS.00044, RQ.NLFS.00035, RQ.NLFS.00036, RQ.NLFS.00037, RQ.NLFS.00038	TBD
UI/LQD-DAQ Integration Test	RQ.NLFS.00012, RQ.NLFS.00013, RQ.NLFS.00029, RQ.NLFS.00040, RQ.NLFS.00049, RQ.NLFS.00050, RQ.NLFS.00056, RQ.NLFS.00057, RQ.NLFS.00058, FC.NLFS.LD.2, FC.NLFS.LQDDAQ.2	TBD
BOM Component Analysis	RQ.NLFS.00047, RQ.NLFS.00048, RQ.NLFS.00068	TBD
Harnessing Analysis	RQ.NLFS.00051, RQ.NLFS.00062, RQ.NLFS.00064, RQ.NLFS.00065, RQ.NLFS.00070, RQ.NLFS.00071	TBD
Schematic Analysis	RQ.NLFS.00051, RQ.NLFS.00062, RQ.NLFS.00064	TBD
Hardware Error Handling	RQ.NLFS.00052, RQ.NLFS.00053,	TBD

	FC.NLFS.LJ.1, FC.NLFS.LJ.2	
Dashboard Unit Test: hazard-mitigation.test.tsx	FC.NLFS.LD.2, FC.NLFS.LD.3, FC.NLFS.LQDDAQ.1, FC.NLFS.LQDDAQ.2	TBD
LQD-DAQ Unit Test: test_api.py::test_snapshot_returns_500_when_stale	FC.NLFS.LQDDAQ.4	TBD
LQD-DAQ Unit Test: test_api.py::test_actuator_busy_returns_423	FC.NLFS.LQDDAQ.5	TBD

7. TEST READINESS

The project has completed its initial development and is ready to enter the testing phase.

7.1. CHANGES SINCE PREVIOUSLY REVIEWED BASELINE

Significant changes have been made to the design, most notably removing SDEC-API as a component for this release.

Software:
SDEC-API has been postponed from the release plan since this version of the software will only be used on the test cart. Ease-of-use requirements have been postponed due to the need for quick design iteration and the lack of safety impact. Rate limiting has been postponed.

Requirement updates:
Postponed: RQ.NLFS.00041, RQ.NLFS.00015, RQ.NLFS.00016, RQ.NLFS.00017, RQ.NLFS.00018, RQ.NLFS.00019, RQ.NLFS.00020, RQ.NLFS.00021, RQ.NLFS.00022, RQ.NLFS.00023, RQ.NLFS.00024, RQ.NLFS.00025, RQ.NLFS.00026, RQ.NLFS.00031, RQ.NLFS.00032, RQ.NLFS.00033, RQ.NLFS.00034

Hardware:
Flight hardware has been postponed from the release plan since this version of the software will only be used on the test cart. Some unverifiable requirements have been postponed. Requirements related to the enclosure have been postponed.

Requirements postponed:
RQ.NLFS.00045, RQ.NLFS.00046, RQ.NLFS.00054, RQ.NLFS.00055, RQ.NLFS.00059 RQ.NLFS.00060, RQ.NLFS.00061, RQ.NLFS.00063, RQ.NLFS.00066, RQ.NLFS.00072

FHA updates:
Postponed mitigations: (related to flight: FC.NLFS.FD.1, FC.NLFS.GS.1, FC.NLFS.GS.2, FC.NLFS.GS.3, FC.NLFS.LD.1, FC.NLFS.SDEC.1, FC.NLFS.SDEC.2, FC.NLFS.SDEC.3), (never implemented, security risk accepted and mitigated by access control/disabling wifi during tests: FC.NLFS.NI.1, FC.NLFS.NI.2, FC.NLFS.LQDDAQ.3)

7.2. APPROVAL

The project meets the requirements to begin testing:

1. Development is complete and all requirements have been implemented
2. Quality assurance practices have been followed, and an audit has been conducted on a random sample of constituent parts
3. Test procedures have been defined and all requirements will be tested by the procedures

Date: 10/31/2026
Project Engineer: Eli Sells
Quality Assurance: Allison Cheon

8. TEST RESULTS

The project has completed its test procedures and have produced the following results:

9. FLIGHT QUALIFICATION

The project has completed all required testing and has passed with minor or no revisions.

9.1. CHANGES SINCE PREVIOUSLY REVIEWED BASELINE

Updates to automated test procedures to ensure full code coverage. Fixed a bug with the lora preset download feature.

9.2. APPROVAL

The project meets the requirements for flight:

1. Test procedures have been applied and have passed their acceptance criteria
2. A random sample of test procedures shows that they were performed as specified
3. No major design or production revisions have been made since test completion

Date:
Project Engineer: Eli Sells
Quality Assurance:
Chief Engineer:

10. QUALIFIED VERSIONS

VERSION	QC Status
v2.4.0	Flight Qualified - Grandfathered
v2.5.0	Flight Qualified - Grandfathered
v2.6.0	Flight Qualified - Grandfathered
v2.7.0	Pending design review